

CHAPTER-3-

DC & AC BRIDGES

3.1 Introduction to Bridge.

- ✓ Bridge circuits are the instruments for making **comparison measurements**, are widely used to measure **resistance, inductance, capacitance and impedance**.
- ✓ Bridge circuits operate on a **null-indication principle**.. It is very accurate.

3.2 Direct Current Bridge.

3.2.1 The Wheatstone Bridge.

- **Sensitivity of the Wheatstone Bridge.**
- **Unbalance Wheatstone Bridge.**

3.2.2 Kelvin Bridge.

3.2.1 The Wheatstone Bridge.

- ✓ The Wheatstone bridge consists of *two parallel resistance* branches with each branch containing two series resistor elements, **Figure 3.1**.
- ✓ A DC voltage source is connected across the resistance network to provide a source of current through the resistance network.
- ✓ A *null detector* is the galvanometer which is connected between the parallel branches to detect the *balance condition*.
- ✓ The Wheatstone bridge is an *accurate* and *reliable* instrument and heavily used in the industries.

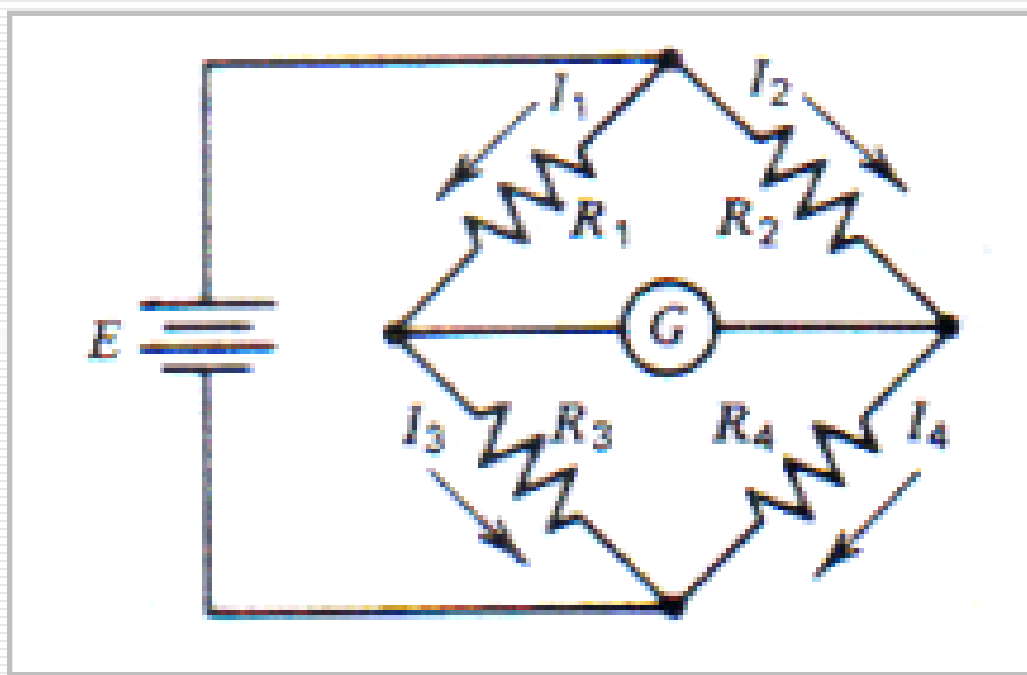


Figure 3.1: Wheatstone Bridge Circuit.

Operation:

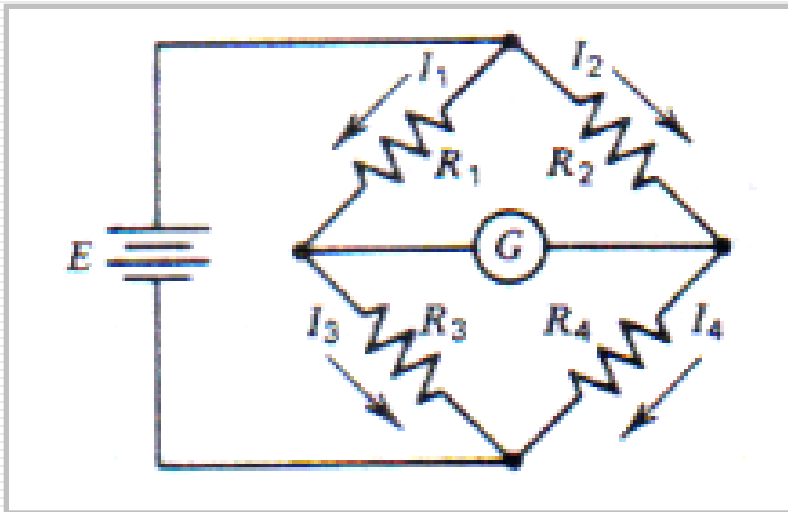
- ✓ We want to know the *value of R_4* , vary one of the remaining resistors until the current through the null detector decreases to *zero*.
- ✓ The bridge is in *balance condition*, the voltage across resistor R_3 is equal to the voltage drop across R_4 , (*$R_3 = R_4$*).

- ✓ At balance the voltage drop at R_1 and R_2 must be equal to.

$$I_3 R_3 = I_4 R_4$$

- ✓ **No current** go through the galvanometer G, the bridge is in balance so,

$$I_1 R_1 = I_2 R_2$$



$$I_1 = I_3$$

$$I_2 = I_4$$

$$I_1 R_3 = I_2 R_4$$

$$\frac{R_1}{R_3} = \frac{R_2}{R_4}$$

or

$$R_1 R_4 = R_2 R_3$$

- ✓ This equation, $R_1 R_4 = R_2 R_3$, states the condition for a balance Wheatstone bridge and can be used to compute the value of unknown resistor.

Example 3.1

Determine the value of unknown resistor, R_x in the circuit of **Figure 3.2** assuming a null exist, current through the galvanometer is zero.

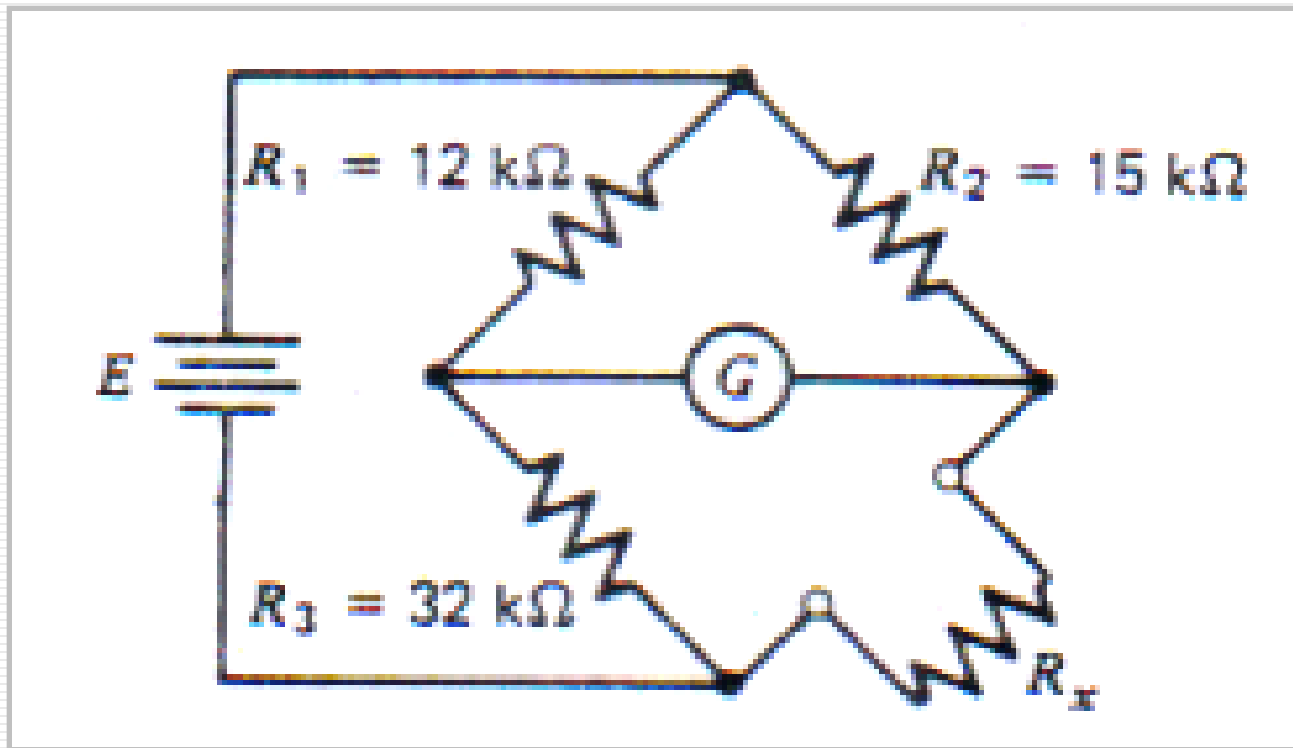


Figure 3.2: Circuit For Example 3.1.

Solution:

From the circuit, the product of the resistance in opposite arms of the bridge is balance, so solving for R_x

$$R_x R_1 = R_2 R_3$$

$$\begin{aligned} R_x &= \frac{R_2 R_3}{R_1} \\ &= \frac{15 K\Omega * 32 K\Omega}{12 K\Omega} = 40 K\Omega \end{aligned}$$

Sensitivity of the Wheatstone Bridge.

- ✓ When the bridge is in *unbalance condition*, current flows through the galvanometer causing a *deflection of its pointer*.
- ✓ The *amount of deflection* is a function of the *sensitivity* of the galvanometer.
- ✓ Sensitivity is the deflection per unit current.
- ✓ The *more sensitive* the galvanometer will *deflect more* with the same amount of current.

$$S = \frac{\text{milimeters}}{\mu A} = \frac{\text{degrees}}{\mu A} = \frac{\text{radian}}{\mu A}$$

- ✓ Total deflection *D* is,

$$D = S * I$$

Unbalanced Wheatstone Bridge.

- ✓ The current flows through the galvanometer can determine by using *Thevenin theorem*.

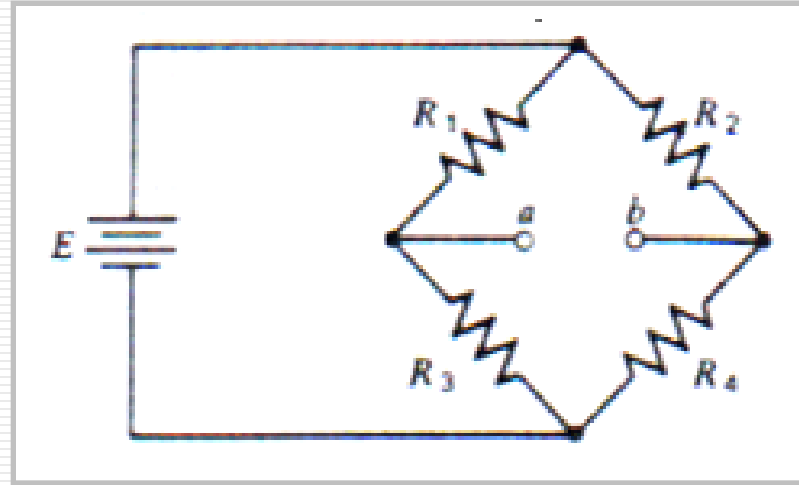
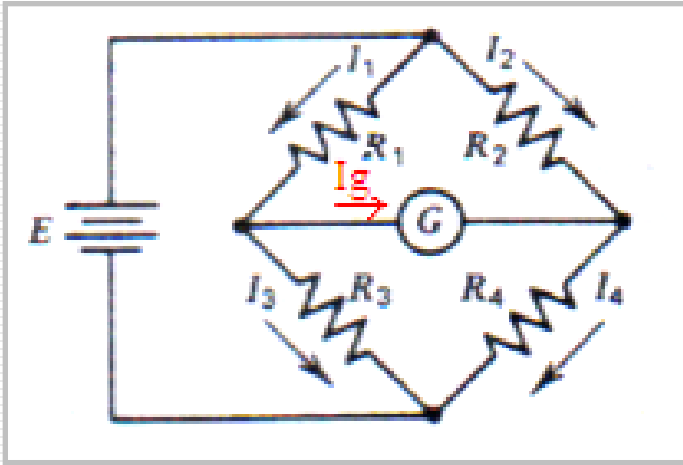


Figure 6.3: Unbalance Wheatstone Bridge.

$$\begin{aligned} R_{Th} &= R_{ab} = (R_1 // R_3) + (R_2 // R_4) \\ &= \frac{R_1 R_3}{R_1 + R_3} + \frac{R_2 R_4}{R_2 + R_4} \end{aligned}$$

$$\begin{aligned} V_{Th} &= V_a - V_b \\ &= E \frac{R_3}{R_1 + R_3} - E \frac{R_4}{R_2 + R_4} \end{aligned}$$

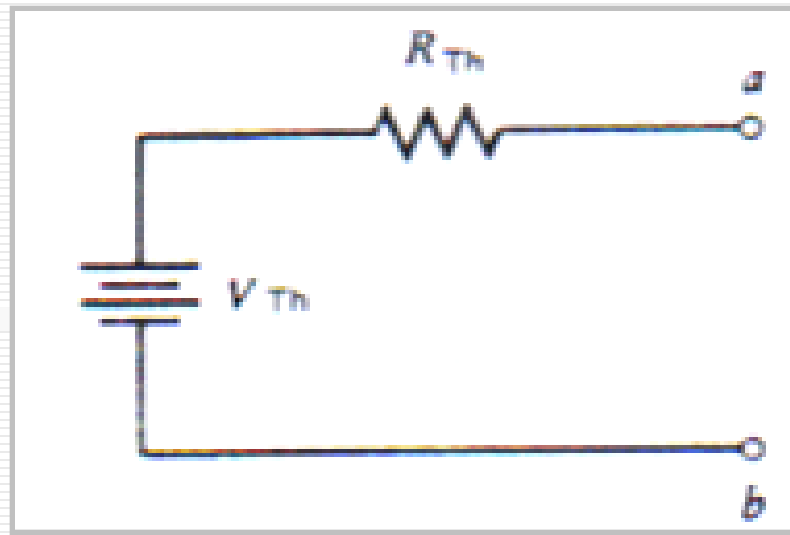


Figure 6.4: Thevenin's Equivalent Circuit for an Unbalanced Wheatstone Bridge.

✓ The deflection current in the galvanometer is,

$$I_g = \frac{V_{th}}{R_{th} + R_g}$$

✓ R_g = the internal resistance in the galvanometer

Example 3.2

Calculate the value of R_x in the circuit of **Figure 3.3** if $V_{Th} = 24 \text{ mV}$ and $I_g = 13.6 \text{ uA}$.

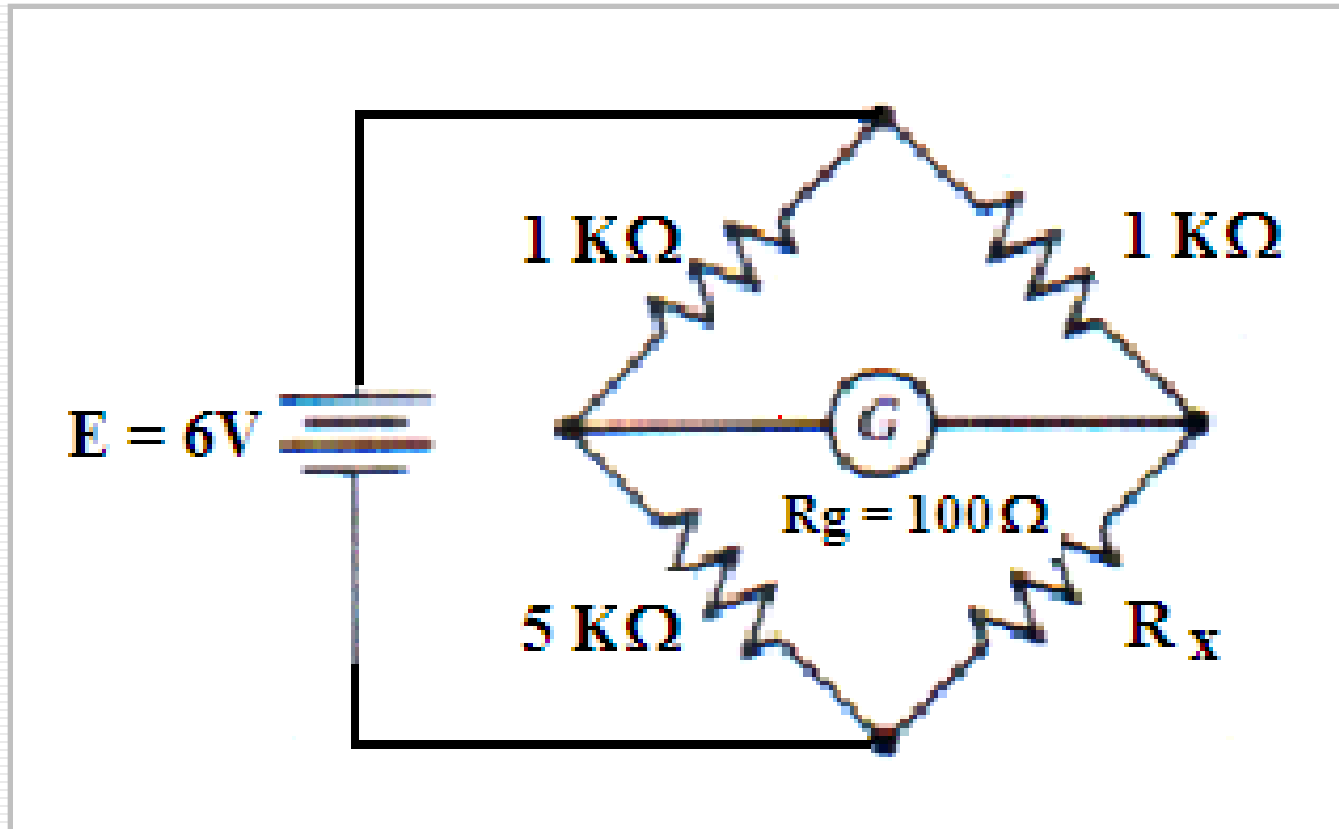


Figure 3.3: Circuit For Example 3.2A.

Solution:

Calculate R_{th}

$$I_g = \frac{V_{th}}{R_{Th} + R_g}$$

$$R_{Th} = \frac{V_{th}}{I_g} - R_g$$

$$R_{Th} = \frac{24mV}{13.6\mu A} - 100\Omega$$

$$R_{Th} = 1.665 K\Omega$$

Calculate R_x

$$R_{Th} = R_{ab} = (R_1 // R_3) + (R_2 // R_x)$$

$$R_{Th} = \frac{R_1 R_3}{R_1 + R_3} + \frac{R_2 R_x}{R_2 + R_x}$$

$$R_{Th} - \frac{R_1 R_3}{R_1 + R_3} = \frac{R_2 R_x}{R_2 + R_x}$$

$$(R_2 + R_x) * \left(R_{Th} - \frac{R_1 R_3}{R_1 + R_3} \right) = R_2 R_x$$

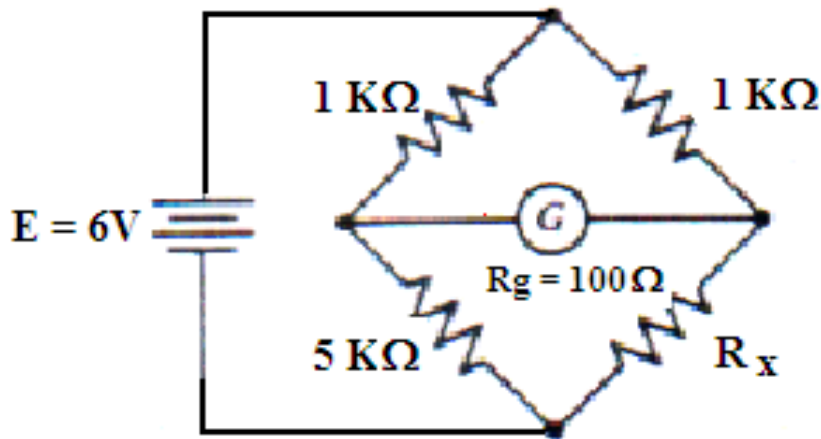
$$\frac{(R_2 + R_x)}{R_x} = \frac{R_2}{\left(R_{Th} - \frac{R_1 R_3}{R_1 + R_3} \right)}$$

$$\frac{R_2}{R_x} = \frac{R_2}{\left(R_{Th} - \frac{R_1 R_3}{R_1 + R_3} \right)} - 1$$

$$R_x = \frac{R_2 \left(R_{Th} - \frac{R_1 R_3}{R_1 + R_3} \right)}{R_2 - \left(R_{Th} - \frac{R_1 R_3}{R_1 + R_3} \right)}$$

$$R_x = \frac{1K\Omega \left(1.665 K\Omega - \frac{1K\Omega * 5K\Omega}{1K\Omega + 5K\Omega} \right)}{1K\Omega - \left(1.665 K\Omega - \frac{1K\Omega * 5K\Omega}{1K\Omega + 5K\Omega} \right)}$$

$$R_x = 4.941 K\Omega$$



Example 3.3:

circuit **Figure 3.4.** Given that $E=6\text{V}$, $R_1=1\text{k}\Omega$, $R_2=1.6\text{k}\Omega$, $R_3=3.5\text{k}\Omega$, $R_4=7.5\text{k}\Omega$ and $R_g=200\Omega$. Find Thevenin equivalent circuit

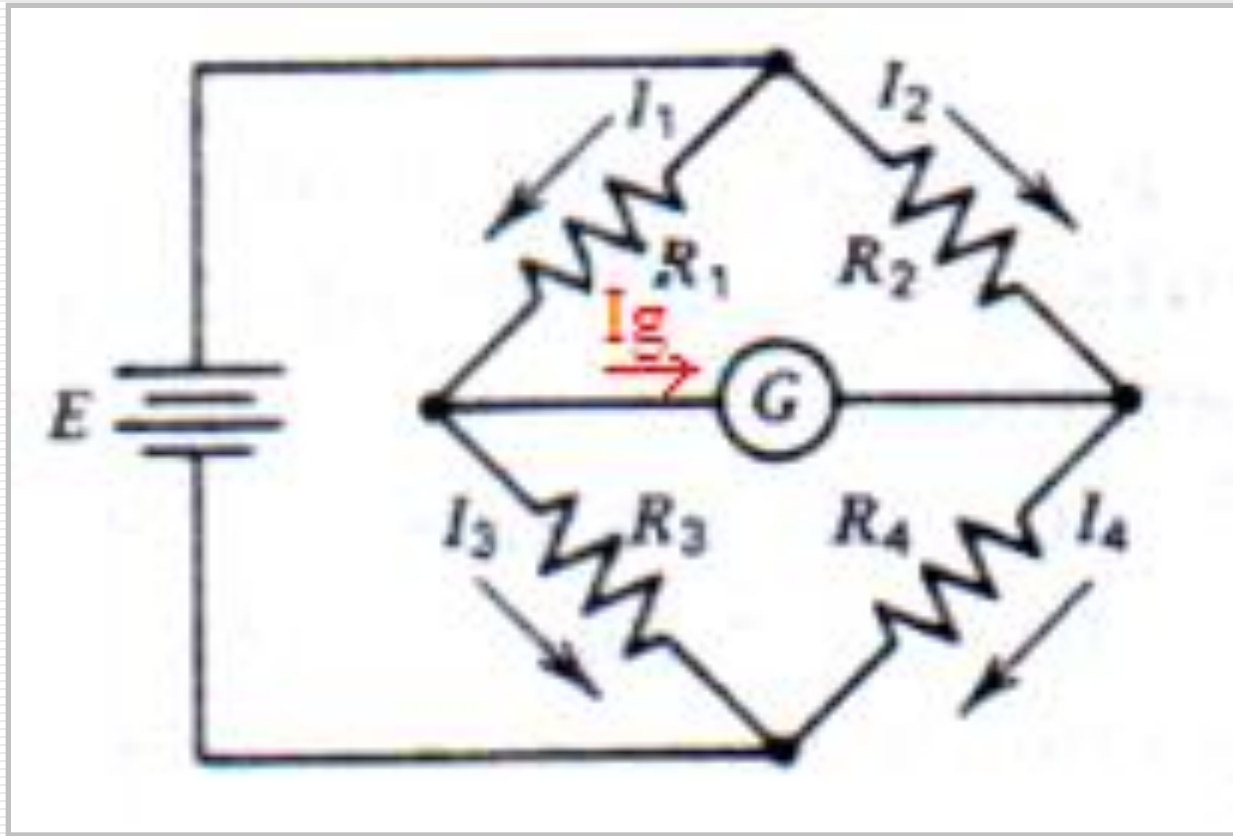


Figure 3.4: Circuit for Example 3.3.

Solution:

(1) Find Thevenin equivalent circuit as seen from by the galvanometer, V_{th} is,

$$\begin{aligned} V_{Th} &= E \left(\frac{R_3}{R_3 + R_1} - \frac{R_4}{R_4 + R_2} \right) \\ &= 6V \left(\frac{3.5K\Omega}{3.5K\Omega + 1K\Omega} - \frac{7.5K\Omega}{7.5K\Omega + 1.6K\Omega} \right) \\ &= 6V(0.778 - 0.824) = 0.276V \end{aligned}$$

(2) Find Thevenin's equivalent resistance (R_{th}) is,

$$\begin{aligned} R_{Th} &= \left(\frac{R_1 R_3}{R_1 + R_3} + \frac{R_2 R_4}{R_2 + R_4} \right) \\ &= 6 \left(\frac{1K\Omega * 3.5K\Omega}{1K\Omega + 3.5K\Omega} + \frac{1.6K\Omega * 7.5K\Omega}{1.6K\Omega + 7.5K\Omega} \right) \\ &= 2.097 K\Omega \end{aligned}$$

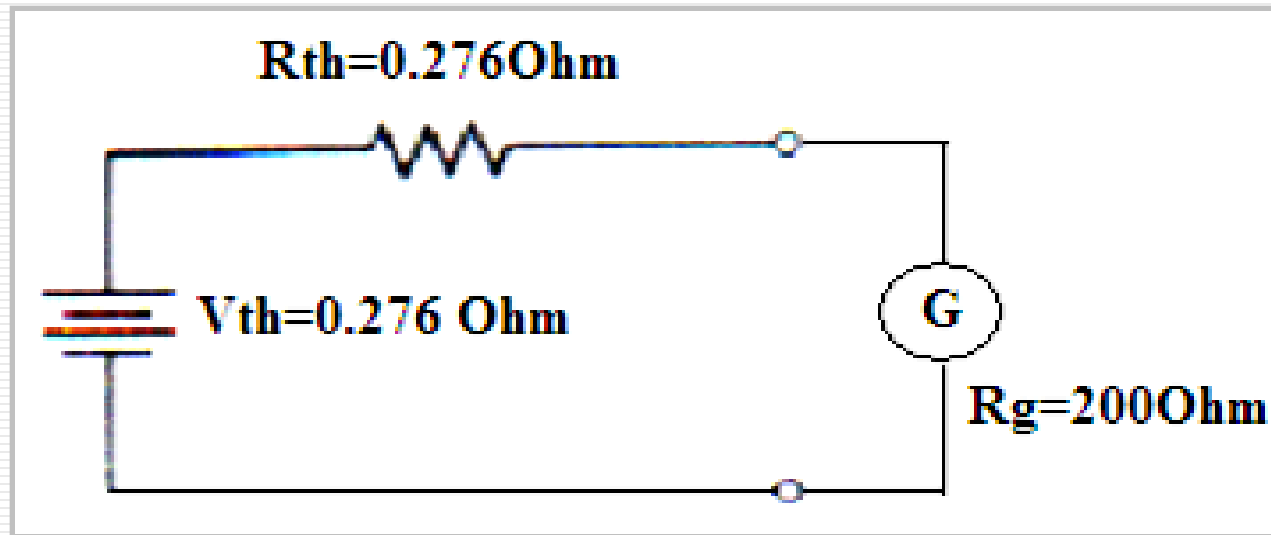
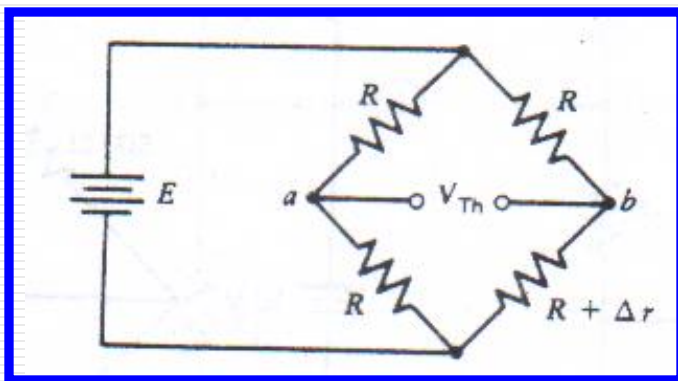


Figure 3.5: Thevenin's Equivalent Circuit for the Example 3.3 Unbalance Bridge.

Slightly Unbalanced Wheatstone Bridge

If three of the four resistors in a bridge are equal to R and the fourth differs by 5% or less, we can develop an approximate expression for Thevenin's equivalent voltage and resistance.

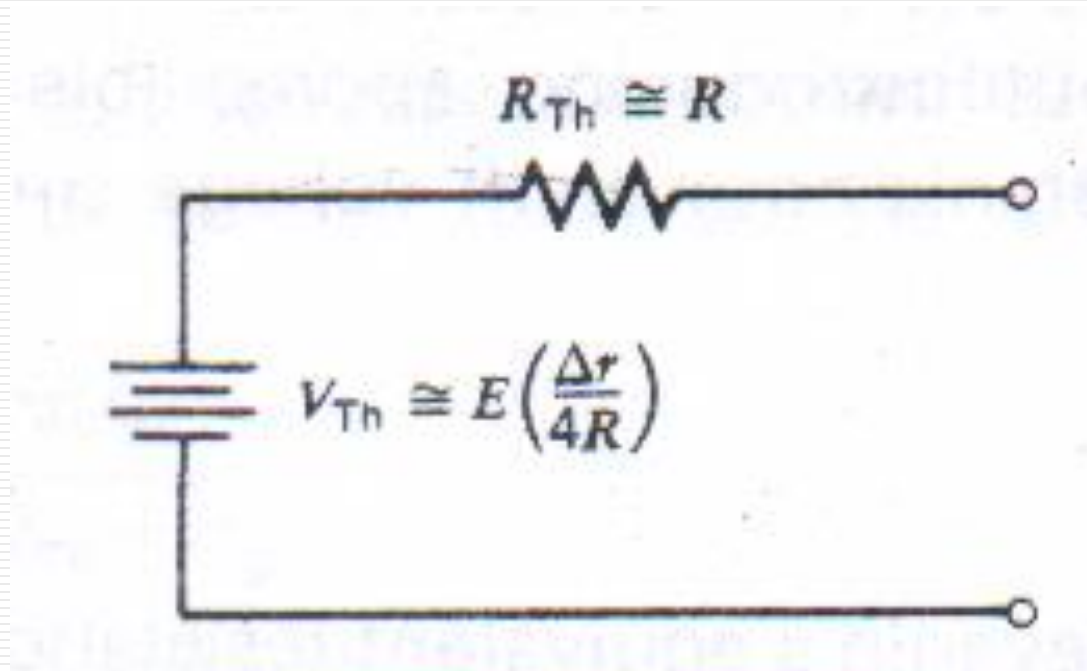


$$V_{th} \approx \frac{E \times \Delta r}{4R} = \left(\frac{\Delta r}{4R} \right) E$$

$$V_{th} = V_b - V_a = \left(\frac{R + \Delta r}{R + R + \Delta r} - \frac{1}{2} \right) \times E = \left(\frac{\Delta r}{4R + 2\Delta r} \right) \times E$$

To find R_{th} :

$$R_{th} = \frac{R}{2} + \frac{R}{2} = R$$



3.6 An approximate Thevenin's equivalent circuit

Example 3.5

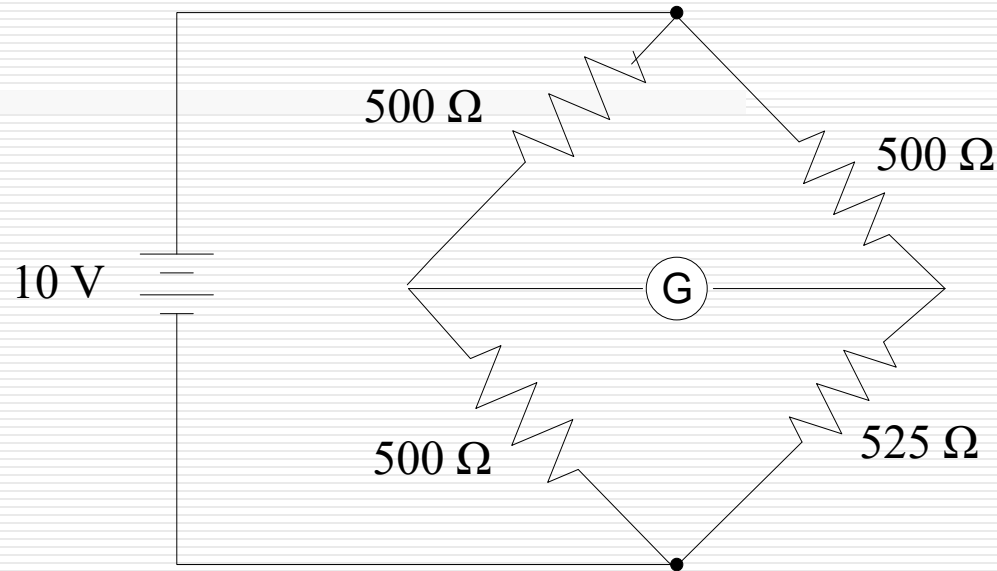


Figure 3.7

Use the approximation equation to calculate the current through the galvanometer in Figure 3.7. The galvanometer resistance, R_g is 125 Ω and is a center zero 200-0-200-μA movement.

Solution:

From formula,

$$I_g = \frac{V_{Th}}{R_{Th} + R_g}$$

(1) Find Thevenin equivalent voltage (V_{th}) is,

$$V_{Th} = E \left(\frac{\Delta r}{4R} \right) = 10V * \frac{25\Omega}{2000\Omega} = 0.125V$$

(2) Find Thevenin equivalent resistance (R_{th}) is,

$$R_{Th} \approx R \approx 500\Omega$$

(3) The current through the galvanometer (I_g) is,

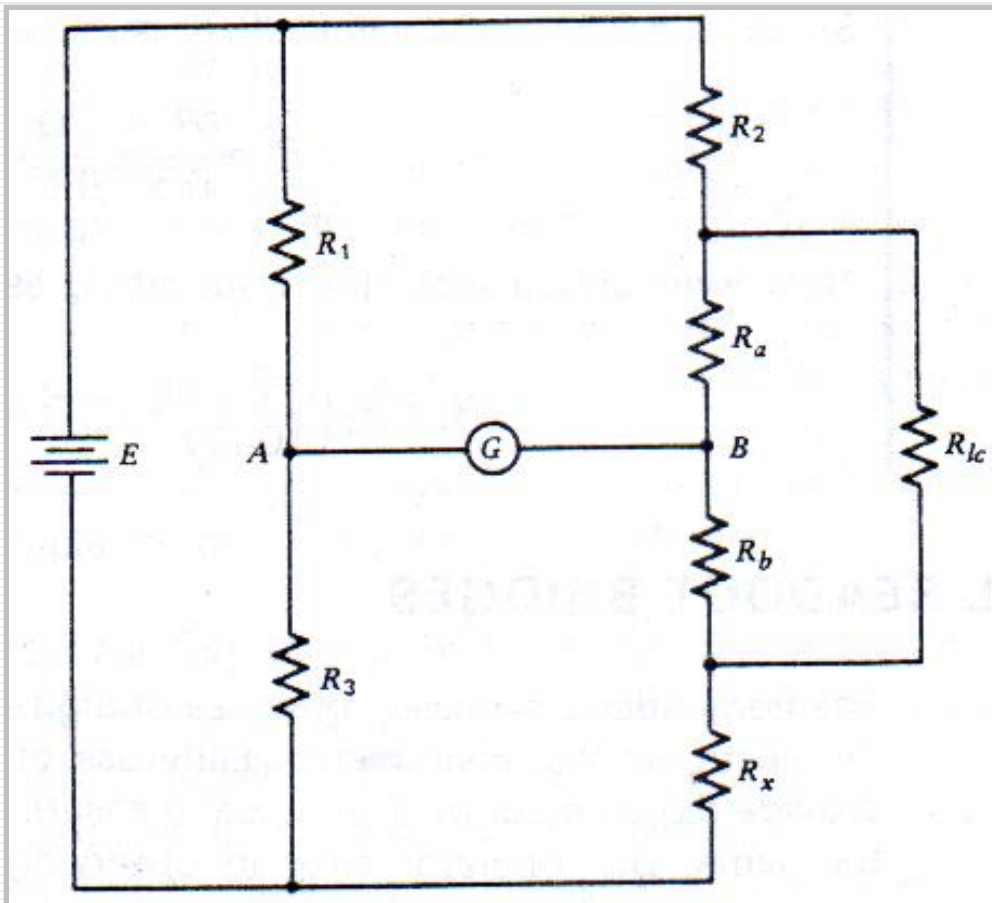
$$I_g = \frac{V_{Th}}{R_{Th} + R_g} = \frac{0.125V}{500\Omega + 125\Omega} \\ = 200 \mu A$$

Observation: If the deflector is a 200-0-200-uA galvanometer, the pointer deflected full scale for a 5% change in resistance.

3.2.2 Kelvin Bridge.

- ✓ The Kelvin Bridge is the *modified version* of the Wheatstone Bridge.
- ✓ The modification is done to *eliminate* the effect of *contact* and *lead resistance* when measuring unknown low resistance.
- ✓ By using Kelvin bridge, resistor within the range of *1 Ω to approximately $1\mu\Omega$* can be measured with high degree of accuracy.
- ✓ **Figure 6.8** is the basic Kelvin bridge. The resistor R_{lc} represent the lead and contact resistance present in the Wheatstone bridge.

- ✓ The second set of R_a and R_b compensates for this relatively low lead contact resistance .
- ✓ At balance the ratio of R_a and R_b must be equal to the ratio of R_1 to R_3 .



$$R_x = \frac{R_2 R_3}{R_1}$$

$$\frac{R_x}{R_2} = \frac{R_3}{R_1}$$

$$\frac{R_x}{R_2} = \frac{R_3}{R_1} = \frac{R_b}{R_a}$$

Figure 6.8: Basic Kelvin Bridge.

Example 3.6

Figure 3.9 is the Kelvin Bridge, the ratio of R_a to R_b is 1000. R_1 is 5 Ohm and $R_1 = 0.5 R_2$. Find the value of R_x .

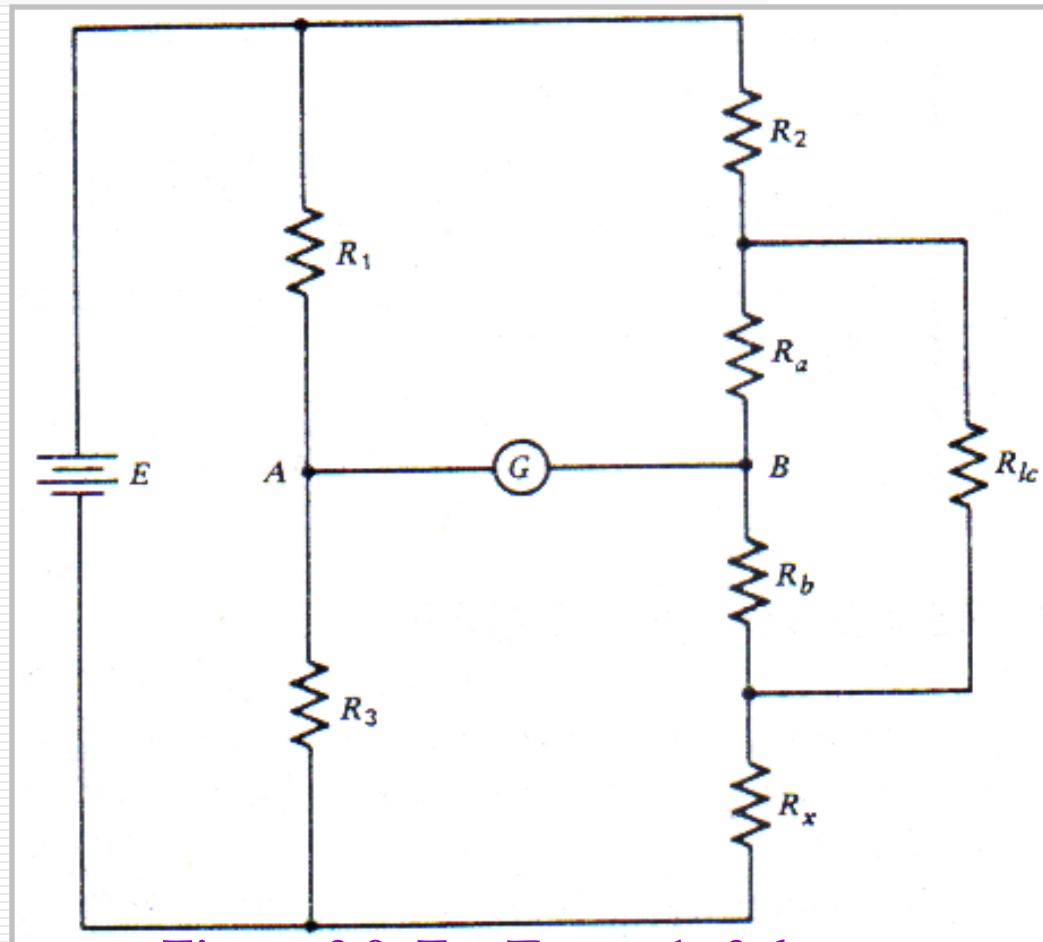


Figure 3.9: For Example 3.6.

Solution:

Calculate the resistance of R_x ,

$$\frac{R_x}{R_2} = \frac{R_b}{R_a} = \frac{1}{1000}$$

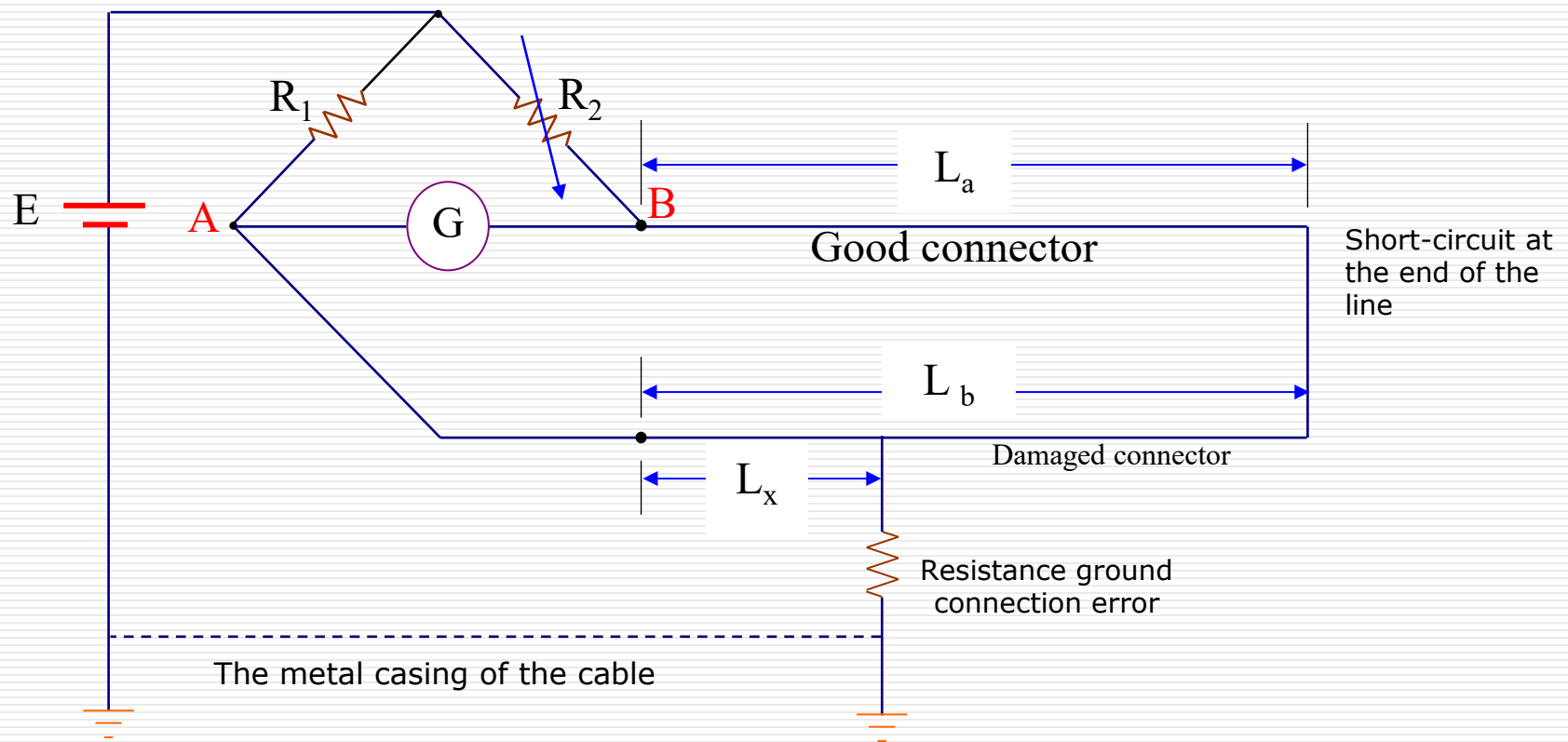
$R_1 = 0.5 R_2$, so calculate R_2

$$R_2 = \frac{R_1}{0.5} = \frac{5\Omega}{0.5} = 10\Omega$$

Calculate the value of R_x

$$\begin{aligned} R_x &= R_2 \left(\frac{1}{1000} \right) \\ &= 10\Omega \left(\frac{1}{1000} \right) = 0.01\Omega \end{aligned}$$

■ Applications on D.C Bridges Murray Loop



$$\frac{R_1}{R_x} = \frac{R_2}{R_a + (R_b - R_x)}$$

$$\therefore R_2 R_x = R_1 R_a + R_1 R_b - R_1 R_x$$

$$\therefore R_2 R_x + R_1 R_x = R_1 R_a + R_1 R_b$$

$$\therefore R_x (R_1 + R_2) = R_1 (R_a + R_b)$$

$$\therefore R_x = \frac{R_1 (R_a + R_b)}{(R_1 + R_2)} \qquad \therefore R = \frac{\rho L}{a}$$

$$\therefore \frac{\rho_x L_x}{a_x} = \frac{R_1}{R_1 + R_2} \left(\frac{\rho_a L_a}{a_a} + \frac{\rho_b L_b}{a_b} \right)$$

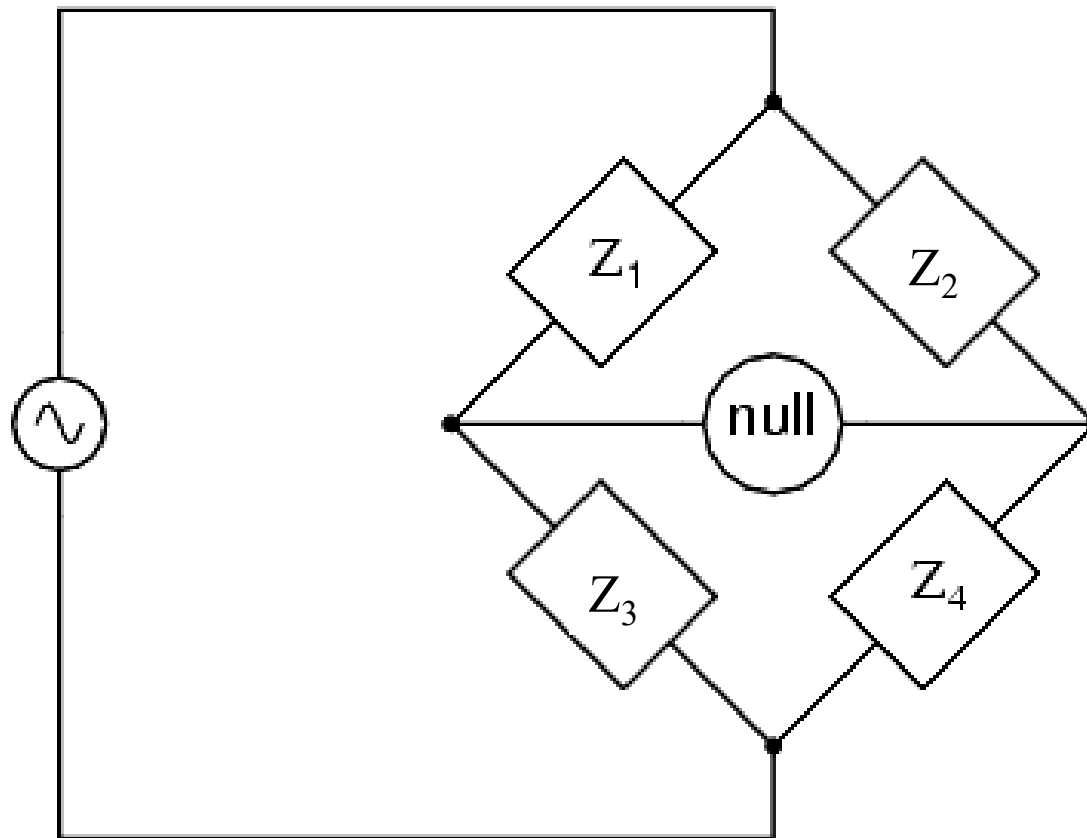
$$\therefore L_x = \frac{R_1}{R_1 + R_2} (2L) = \frac{2R_1 L}{R_1 + R_2}$$

3.3 Alternating Current Bridge.

3.3.1 Introduction to AC Bridge.

- ☑ AC bridges are used to *measure inductance and capacitance*.
- ☑ All the AC bridges are *based on the Wheatstone bridge*.
- ☑ In the AC bridge the bridge circuit consists of *four impedances* and an *ac voltage source*.
- ☑ The impedances can either be *pure resistance* or *complex impedance*.

General impedance bridge



A box with a "Z" written inside is the symbol for any nonspecific impedance.

Figure 3.10: General AC Bridge Circuit.

Operation of AC Bridge:

- ✓ When the specific circuit conditions apply, the *detector current becomes zero*, which is known as *null* or *balance condition*.
- ✓ Since zero current, it means that there is *no voltage difference across the detector*, Figure 3.10.
- ✓ Voltage at point *b* and *c* are equal.
$$I_1 Z_1 = I_2 Z_2$$
- ✓ The same thing at point *d*.
$$I_1 Z_3 = I_2 Z_4$$
- ✓ From two above equation yield *general bridge equation*;

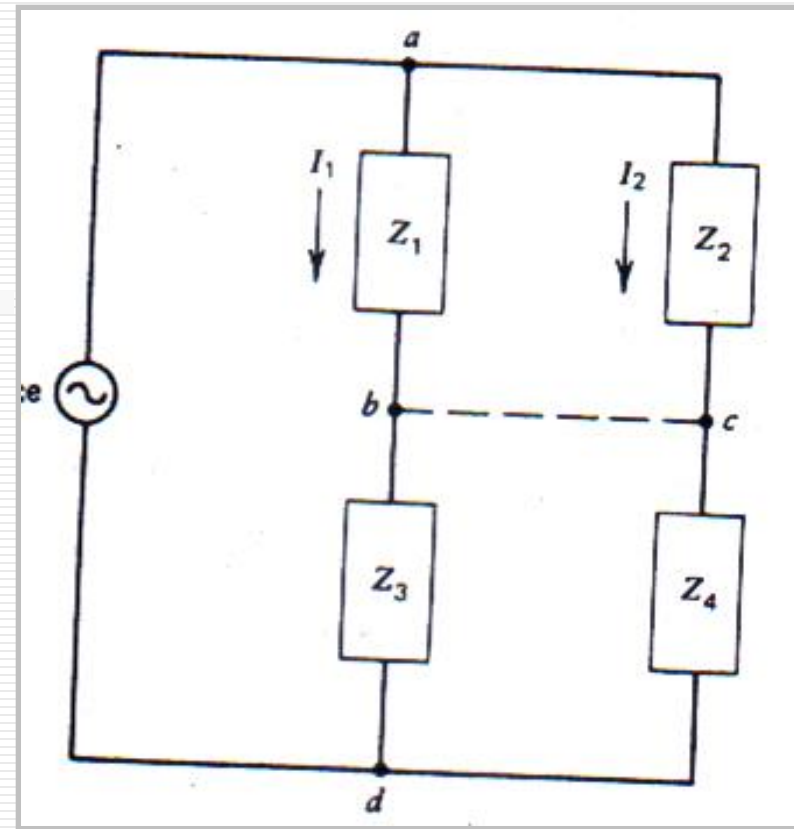


Figure 3.10: Equivalent of Balance (nulled) AC Bridge.

$$\frac{Z_1}{Z_3} = \frac{Z_2}{Z_4}$$

✓ Figure 3. 11(a) and 3.11 (b) is a simple AC Bridge circuit.

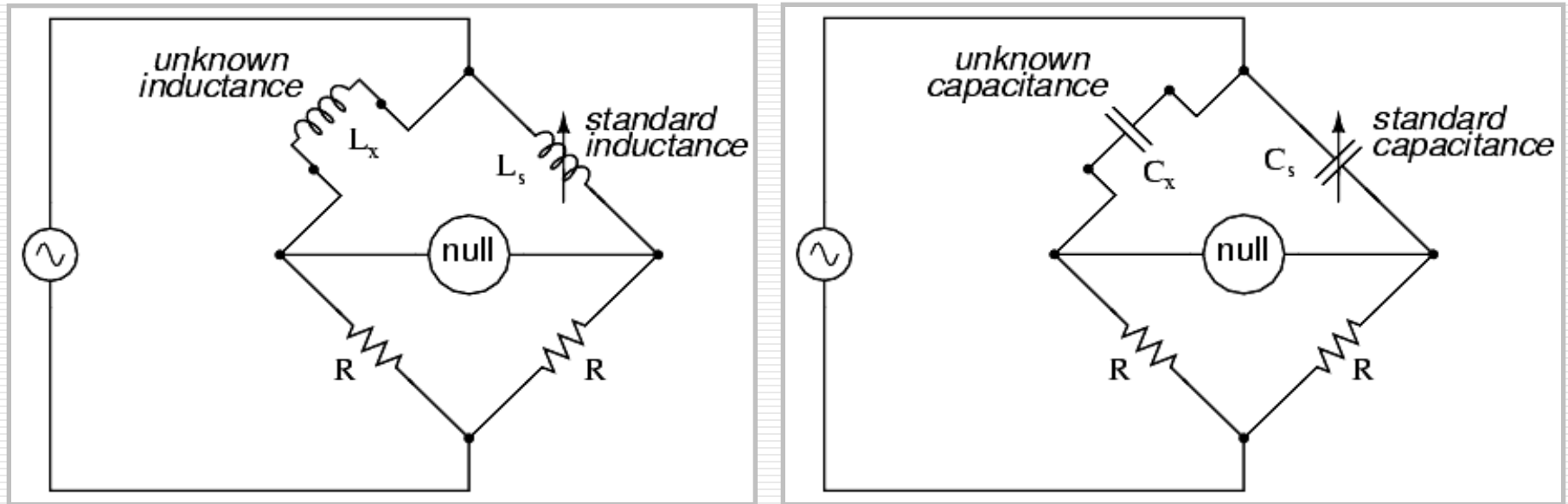


Figure 3.11: (a) and (b) are Simple AC Bridge Circuit.

Example 3.7:

The impedances of the AC bridge in **Figure 3.7** are given as follows,

$$Z_1 = 200 \angle 30^\circ \Omega$$

$$Z_2 = 150 \angle 0^\circ \Omega$$

$$Z_3 = 250 \angle -40^\circ \Omega$$

$$Z_x = Z_4 = \textit{unknown}$$

Determine the constants of the unknown arm.

Solution:

The first condition for bridge balance requires that

$$Z_1 Z_x = Z_2 Z_3$$

$$Z_x = (Z_2 Z_3 / Z_1)$$

$$= [(150 * 250) / 200]$$

$$= 187.5 \Omega$$

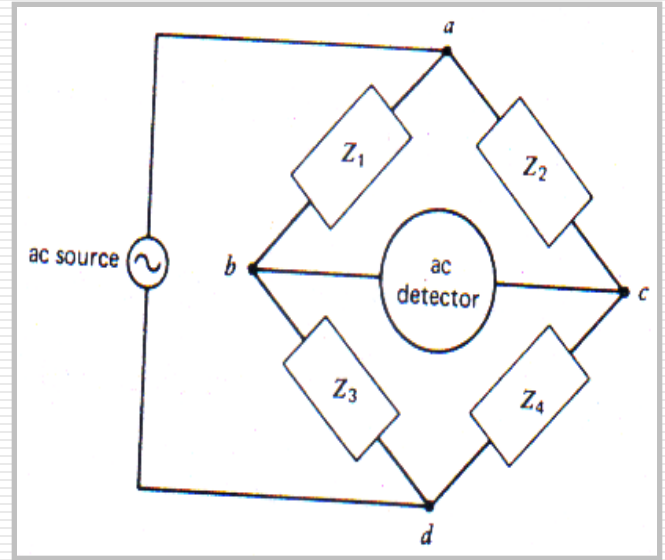


Figure 3.12: Circuit For Example 3.7.

- ✓ The second condition for balance requires that the sums of the phase angles of opposite arms be equal,

$$\begin{aligned}\theta_1 + \theta_x &= \theta_2 + \theta_3 \\ \theta_x &= \theta_2 + \theta_3 - \theta_1 \\ &= 0 + (-40^\circ) - 30^\circ \\ &= -70^\circ\end{aligned}$$

- ✓ The unknown impedance Z_x , can be written as,

$$\begin{aligned}Z_x &= 187.5 \, \Omega / -70 \\ &= (64.13 - j176.19) \, \Omega\end{aligned}$$

- ✓ This indicate that we are dealing with a capacitive element, possibly consisting of a series resistor and a capacitor.

3.3.2 Similar-Angle Bridge

- ✓ **Figure 3.13** is a simple form of Similar–Angle Bridge, which is used to measure the **impedance of a capacitive circuit**.
- ✓ Sometimes called the capacitance *comparison bridge* or series *resistance capacitance bridge*.

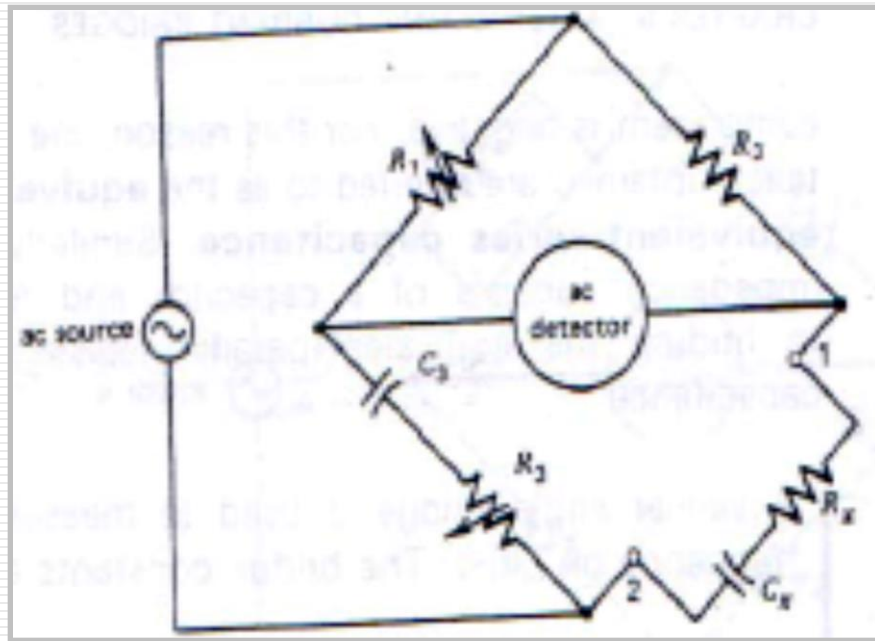


Figure 3.13: Similar-Angle Bridge.

- ✓ The impedance of the arm can be written,

$$Z_1 = R_1$$

$$Z_2 = R_2$$

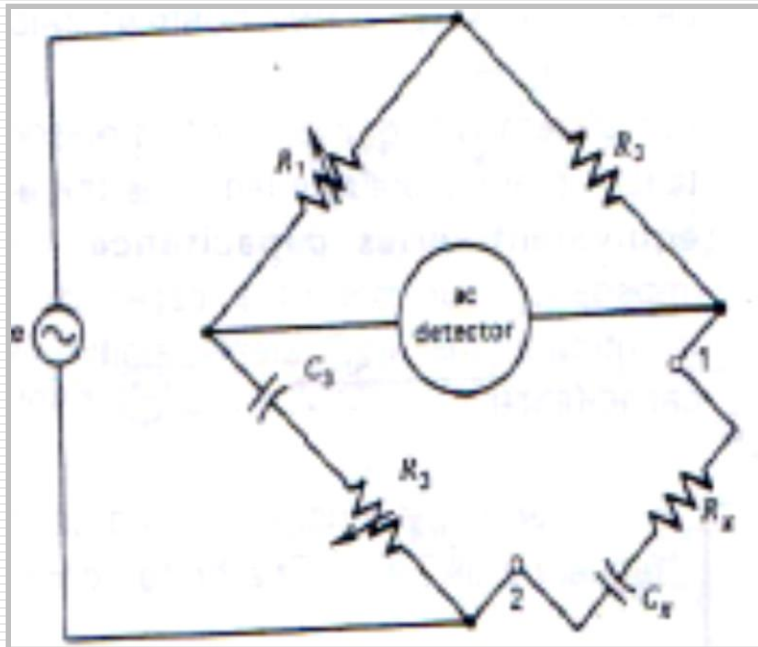
$$Z_3 = R_3 - jX_{c3}$$

$$Z_4 = R_x - jX_{cx}$$

- ✓ Substitute in the balance equation,

$$R_1(R_x - jX_{cx}) = (R_3 - jX_{c3})R_2$$

- ✓ Further simplification,



$$R_1 R_x - jR_1 X_{cx} = R_2 R_3 - jR_2 X_{c3}$$

$$R_1 R_x = R_2 R_3$$

$$-jR_1 X_{cx} = -jR_2 X_{c3}$$

$$-jR_1 \frac{1}{\omega C_x} = -jR_2 \frac{1}{\omega C_3}$$

$$R_1 C_3 = R_2 C_x$$

$$R_x = \frac{R_2}{R_1} R_3$$

$$C_x = \frac{R_1}{R_2} C_3$$

3.3.3 Maxwell-Wein Bridge

- ✓ It is used to measure *unknown inductances* with capacitance standard.
- ✓ Because the phase shifts of inductors and capacitors are exactly opposite each other, a capacitive impedance can balance out an inductive impedance if they are located in opposite legs of a bridge

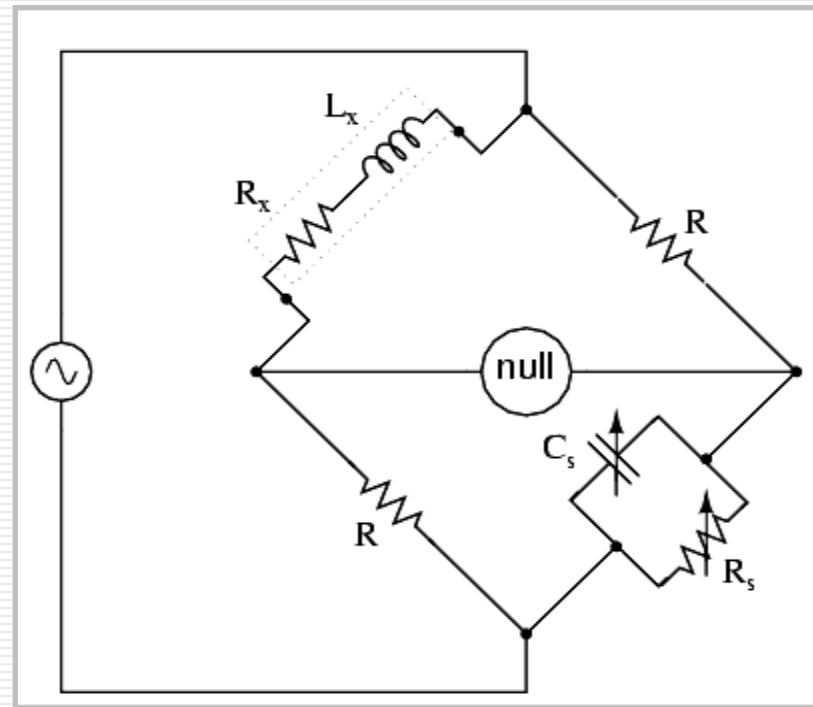


Figure 3.13: Maxwell-Wein Bridge.

✓ **Figure 7.6** is the Maxwell-Wein Bridge or sometimes called a Maxwell bridge.

✓ The *impedance* of the arm can be written as,

$$\begin{aligned}Z_1 &= R_x + jX_{LX} \\Z_2 &= R_2 \\Z_3 &= R_3 \\Z_4 &= \frac{1}{1/R_c + j\omega C_1}\end{aligned}$$

✓ Substitute in the balance equation,

✓ Set real and imaginary part to zero,

$$R_x = \frac{R_2 R_3}{R_1}$$

$$\begin{aligned}\frac{1}{1/R_1 + j\omega C_1} (R_x + jX_{LX}) &= R_2 R_3 \\R_x + j\omega L_x &= \frac{R_2 R_3}{R_1} + j\omega R_2 R_3 C_1 \\L_x &= R_2 R_3 C_1\end{aligned}$$